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I. Channel Protein

Channel protein forms a channel or tunnel in the cell membrane for the easy passage of molecules to enter the cell. The channels are either open or remain closed. They may open up for specific molecules. Some channel proteins create larger pores in the outer membrane. Examples: Porin and Aquaporin.

i. Porin

Porin is a large transporter protein found in the outer membrane of plastids, mitochondria and bacteria which facilitates smaller molecules to pass through the membrane.

ii. Aquaporin

Aquaporin is a water channel protein embedded in the plasma membrane. It regulates the massive amount of water transport across the membrane (Figure 11.3). Plants contain a variety of

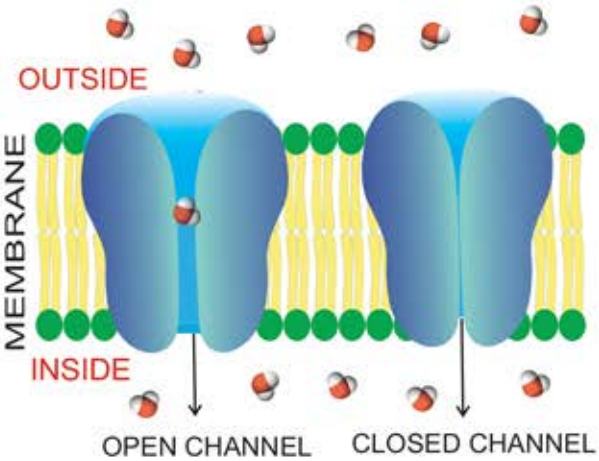


Figure 11.3: Aquaporin

aquaporins. Over 30 types of aquaporins are known from maize. Currently, they are also recognized to transport substrates like glycerol, urea, CO_2, NH_3, metalloids, and **Reactive Oxygen Species** (ROS) in addition to water. They increase the permeability of the membrane to water. They confer drought and salt stress tolerance.



Discovery of Aquaporin



Peter Agre, discovered the "Water Pore" Aquaporin in RBC and received Nobel Prize for chemistry in 2003.

II. Carrier Protein

Carrier protein acts as a vehicle to carry molecules from outside of the membrane to inside the cell and vice versa (Figure 11.4). Due to association with molecules to be transported, the structure of carrier protein gets modified until the dissociation of the molecules.

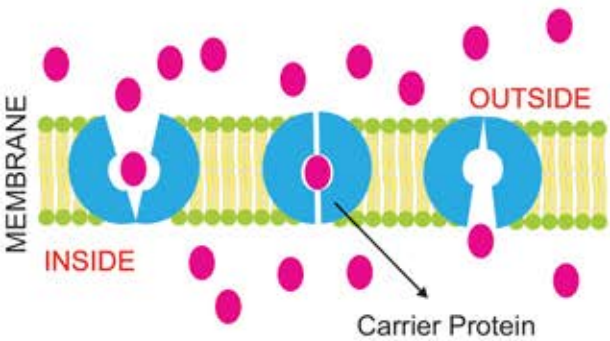


Figure 11.4: Carrier Protein

There are 3 types of carrier proteins classified on the basis of handling of molecules and direction of transport (Figure 11.5). They are, i) **Uniport** ii) **Symport** iii) **Antiport.**

i. ***Uniport***: In this molecule of a single type move across a membrane independent of other molecules in one direction.

ii. ***Symport or co-transport***: The term **symport** is used to denote an integral membrane protein that simultaneously